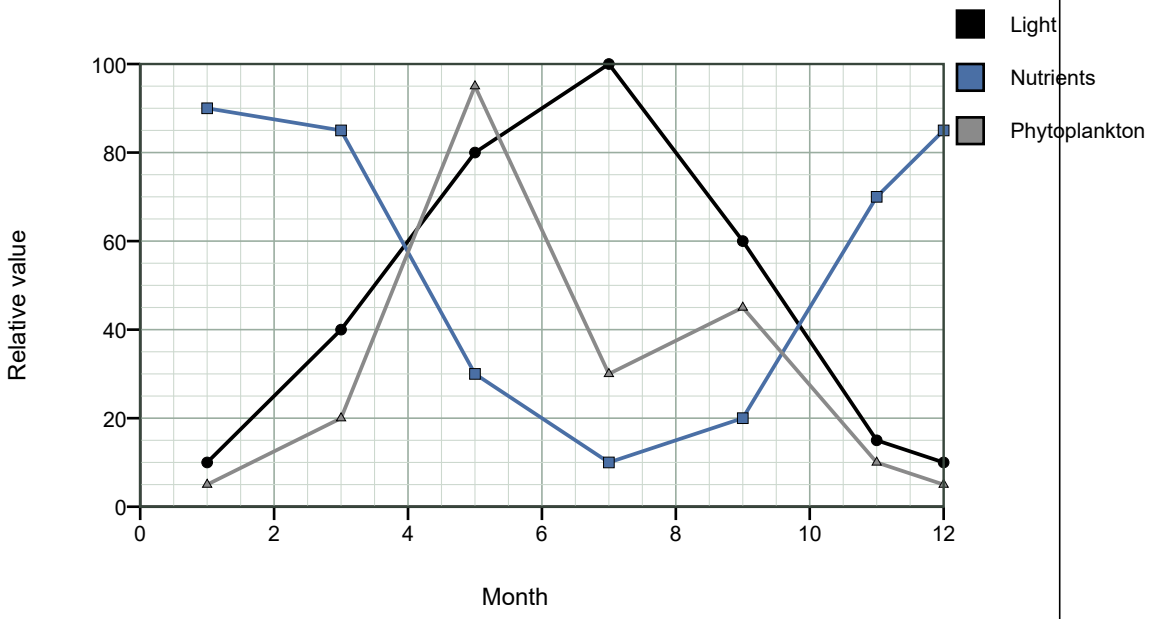
# Questions by topic: Marine productivity & fishing - Set 1 (75 marks)

**Answer all questions in the spaces provided.**

**0 1** Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

**Figure 1**




**0 1 . 1** Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

**[4 marks]**

**0 1 . 2** Explain why tropical open oceans have low productivity all year despite high light and temperature.

**[2 marks]**

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**0 1 . 3** Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

**[3 marks]**

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**0 1 . 4** Explain why phosphate is often the limiting nutrient in the sea.

**[1 mark]**

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**10**

**0 2** Table 1 shows features of some fishing methods.

**0 2 . 1** Complete Table 1.

**[6 marks]**

**Table 1**

| Method | Target species | One environmental disadvantage | |
|---|---|---|---|
| Demersal trawling | | | |
| Purse seining | | | |
| Pelagic long-lining | | | |

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**0 2 . 2** Explain why by-catch is a particular problem with shrimp trawling and drift netting.

**[2 marks]**

**0 2 . 3** Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

**[2 marks]**

**10**

**0 3** Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

**Table 1**

| Species | Age at catchable size / years | Age at first breeding / years | Eggs per female per year | Maximum lifespan / years |
|---|---|---|---|---|
| Atlantic cod | 2 | 4 | Millions | 25 |
| Orange roughy | 20 | 30 | Thousands | 150 |
| Herring | 2 | 2 | Tens of thousands | 20 |

**0 3 . 1** Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

**[4 marks]**

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**0 3 . 2** Explain why herring populations recover from overfishing much faster than cod.

**[2 marks]**

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**0 3 . 3** Explain why the total mass of fish landed can stay the same while a population is being overfished.

**[2 marks]**

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**0 3 . 4** Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

**[2 marks]**

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**0 4** The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where $S_1$ and $S_2$ are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

**Table 1**

| Term | Biomass / thousand tonnes |
|---|---|
| $S_1$ | 600 |
| A | 90 |
| G | 150 |
| C | 220 |
| M | 80 |

**0 4 . 1** Calculate $S_2$ and state whether the stock is being fished sustainably.

**[2 marks]**

**Answer** ☐ thousand tonnes

**0 4 . 2** Calculate the maximum catch that would have kept the stock constant.

**[1 mark]**

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**Answer** [ ] thousand tonnes

**0 4 . 3** Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

**[3 marks]**

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**0 4 . 4** Describe how each of the following measures obtained from commercial catches can indicate overfishing.

**[4 marks]**

**Catch per unit fishing effort** .....

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**Mean fish size and mean age** .....

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**10**

**0 5 . 1** Explain how each of the following reduces the environmental impact of fishing.

**[8 marks]**

**Square-mesh escape panels** .....

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**Minimum and maximum catchable sizes** .....

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**No-take zones** .....

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**Closed seasons** .....

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**0 5 . 2** Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

**[2 marks]**

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**0 6 . 1** Describe **three** methods of reducing seabird by-catch in long-line fisheries.

**[3 marks]**

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**2** .....

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**0 6 . 2** Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

**[3 marks]**

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**0 6 . 3** Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

**[2 marks]**

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**0 6 . 4** Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

**[2 marks]**

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**0 7** Table 1 shows the catch made by a shrimp trawler during one fishing trip.

**Table 1**

| Component of catch | Mass / tonnes |
|---|---|
| Shrimps (target species) | 0.6 |
| Crabs, sea urchins, molluscs and starfish | 5.9 |
| Immature fish of commercial species | 1.9 |

**0 7 . 1** Calculate the percentage of the total catch that was by-catch.

**[2 marks]**

**Answer** ____ %

**0 7 . 2** Explain why shrimp trawling produces such a high proportion of by-catch.

**[1 mark]**

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**0 7 . 3** Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

**[3 marks]**

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**0 7 . 4** Evaluate the methods that can be used to reduce the environmental impacts of fishing.

**[9 marks]**

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**END OF QUESTIONS**

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